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Core/Pan Analysis Reflection

1. Relevance of Metrics

The statistical summary that Panaroo generated provides further important insight into the pangenome diversity and structure. For this dataset, 6598 total genes were identified, with 3686 of these genes being classified as core genes (99-100% present within the genomes). This large amount of core genomes could suggest higher levels of conservation amongst the bacteria strains selected, and may indicate that they are closely related and share multiple essential functions. In comparison there were only 1360 shell genes (somewhat conserved) and 1552 cloud genes (rare genes to be present in a few genomes), this reflects the variability of genetics and possible strain specific adaptation.

The soft core genes (95-99%) absence implies a rather clear gene distinction present and more variable. Suggesting that the dataset is consistent in terms of quality and composition, containing fewer affected borderline genes or inconsistent assemblies. These crucial metrics directly influence the downstream analyses (construction of phylogenic trees and comparison of present/absent genes). For example, having a large and well-defined core genome implies the ability to improve multiple sequence alignment reliability and evolutionary inference. While the accessory genome (shall and cloud genes) demonstrates the diverse functionality and environmental adaptions.

Overall, these results show the combination for high quality genome selection (through BUSCO) and the considerate annotation (through Bakta) to produce this unique pangenome dataset for analysis. This Panaroo output (txt) of core and accessory genes supports the interpretation while minimizing the poor-quality annotation noise that occurs.